



05 OCTOBER 2024

SOFTWARE TESTING LAB

JAVA CODE'S TESTING

AMJAAD HAWSAWI 444003283
WEAAM RASHED 444015412
NOORAH AL-KABKABI 444003676



Experiment1:

Q#1- Write a test case to validate the system behavior against entering empty username and password.

```
1 import java.util.*;
2 import java.lang.*;
3 import java.io.*;
4
5 class Codechef
6 {
7     public static class UserRegistrationTest {
8         public static void main(String[] args) {
9             UserRegistration userReg = new UserRegistration();
10
11             boolean result = userReg.registerUser("", "", "test@example.com");
12
13             System.out.println("Test Case TC001 - Expected: false, Actual: " + result);
14         }
15     }
16 }
17
18 public static class UserRegistration {
19     public boolean registerUser(String username, String password, String email) {
20         if (username == null || password == null || email == null) {
21             return false;
22         }
23         if (username.isEmpty() || password.isEmpty() || email.isEmpty()) {
24             return false;
25         }
26         if (!email.contains("@") || !email.contains(".")) {
27             return false;
28         }
29         // simulate sending confirmation email
30         System.out.println("Confirmation email sent to: " + email);
31         return true;
32     }
33 }
34 }
35 }
36 }
```

Test Case:

Test ID: TC-REG-01

Title: verify system do not allow registration with missing data

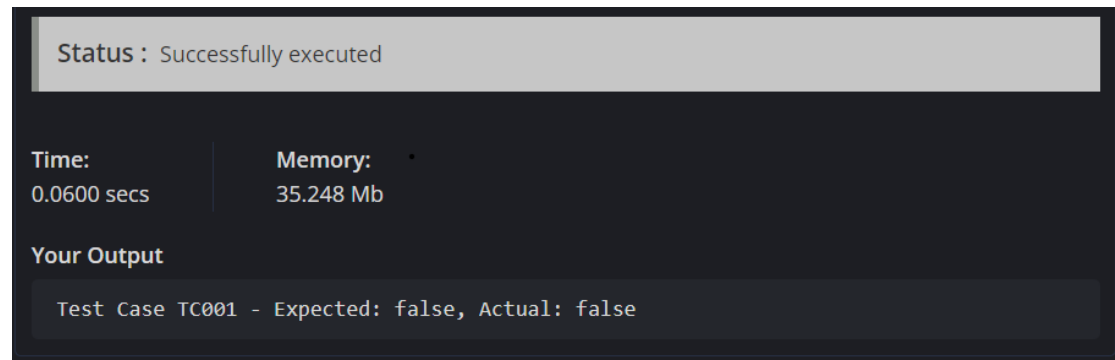
Steps: 1-Initialize the UserRegistration class

2-Call the registerUser method with empty strings for username and password, and a valid email

3-Observe the result

Expected Result: false

Actual Result: false



The screenshot shows a test execution interface. At the top, a grey bar indicates 'Status : Successfully executed'. Below this, a table displays 'Time: 0.0600 secs' and 'Memory: 35.248 Mb'. Under the 'Your Output' section, the text 'Test Case TC001 - Expected: false, Actual: false' is shown.

Time:	Memory:
0.0600 secs	35.248 Mb

Your Output

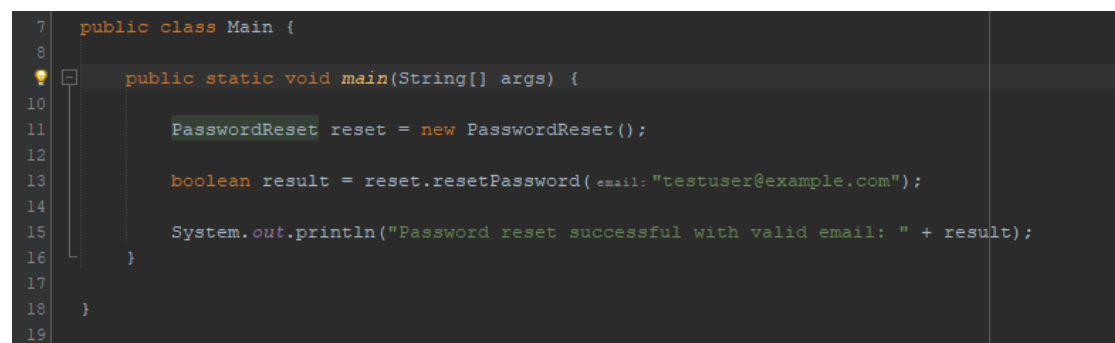
Test Case TC001 - Expected: false, Actual: false

Q#2- Write a test Case to verify the successful user password reset with valid email written.

Test Case:

Test ID: TC-PWD-NEG-01

Title: Verify successful user password reset with valid email.



```
7 public class Main {
8
9     public static void main(String[] args) {
10
11         PasswordReset reset = new PasswordReset();
12
13         boolean result = reset.resetPassword(email: "testuser@example.com");
14
15         System.out.println("Password reset successful with valid email: " + result);
16     }
17
18 }
19
```

Steps:

- 1- Creating an instance of PasswordReset class.
- 2 - Call the resetPassword method with a valid email.
- 3 – Print the result.

Expected Result: The system should return true.

Actual Result: The system returns true as expected.

```
run:
Password reset email sent to: testuser@example.com
Password reset successful with valid email: true
BUILD SUCCESSFUL (total time: 0 seconds)
```

Experiment2:

```
11 public class UserRegistrationWithBug {
12     public boolean registerUser(String username, String password, String email) {
13         if (username == null || email == null) {
14             return false;
15         }
16         if (!email.contains("@") || !email.contains(".")) {
17             return false;
18         }
19         // Simulate sending confirmation email
20         System.out.println("Confirmation email sent to: " + email);
21         return true;
22     }
23 }
```

```

14
15 public static void main(String[] args) {
16
17     UserRegistrationWithBug reg2 = new UserRegistrationWithBug();
18     boolean result3 = reg2.registerUser("testuser", "password123",
19     "testuser@uqucom");
20     System.out.println("Registration successful: " + result3);
21 }
22 }

```

Q#3- Apply the previous code and fill the following, then realize the result and mention if any problem occur and why.

Test Case:

Test ID: TC-REG-NEG-01

Title: Verify that the system does not allow registration with an invalid email.

Steps: call the registerUser("testuser", "password123 testuser@uqu.com") method.

Expected Result: : The system should not register the user and should return false if the email syntax is incorrect.

Actual Result:

-If the email syntax is correct:

```

--- exec:3.1.0:exec (default-cli) @ roaaCode ---
Confirmation email sent to: testuser@uqu.com
Registration successful: true
-----
BUILD SUCCESS
-----

Total time: 2.081 s
Finished at: 2024-10-06T00:39:02+03:00

```

- If the email syntax is incorrect :

```
--- exec:3.1.0:exec (default-cli) @ roaa
Registration successful: false
-----
BUILD SUCCESS
-----
Total time: 2.109 s
Finished at: 2024-10-08T05:04:23+03:00
```

-Which bring us to the conclusion that the previous code is logically correct, and the function "UserRegistrationWithBugs" actually prints a correct output based on the input syntax that we passed to it.